The translated-cone lattice count in every dimension

Solution packet for arXiv:2403.10350

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Abstract

Aksentijević, Aleksić, and Pilipović ask whether two cones in good position satisfy the translated-intersection bound needed for their product theorem in dimension $d \geq 3$, conjecturing exponent d. We prove the bound, with an explicit constant, whenever the closed spherical traces of the cones are antipodally separated. This includes the natural closed-cone interpretation of their hypothesis and works in every dimension. The exponent d is optimal. We also resolve an important terminology ambiguity: if "cone" means open cone and only literal disjointness is imposed, the claim is false even in dimension two, with one translated intersection containing infinitely many lattice points.

1 Source question

For cones $\Gamma_1, \Gamma_2 \subset \mathbb{R}^d \setminus \{0\}$, condition (a) of Theorem 5.5 in the source asks for constants $C > 0$ and $\gamma \geq 1$ such that

$$N(n) := \operatorname{card}\{k \in \mathbb{Z}^d : n - k \in \Gamma_1,\ k \in \Gamma_2\} \leq C|n|^\gamma, \qquad n \in \mathbb{Z}^d. \tag{1}$$

Remark 2 on page 13 of the arXiv PDF says that $\Gamma_1 \cap (-\Gamma_2) = \varnothing$ yields (1) for $d = 2$ with $\gamma = 2$, and hypothesizes that it yields the bound for $d \geq 3$ with $\gamma = d$. The page is reproduced in Figure 1.

The source uses "open convex cone" earlier, but its planar illustration in Remark 2 uses non-strict inequalities. The distinction matters. We therefore state the exact geometric hypothesis, prove the intended result, and then treat the literal open-cone reading.

2 Proof intuition

Every counted k gives a decomposition $n = x + y$ with $x = n - k \in \Gamma_1$ and $y = k \in \Gamma_2$. If the directions of x and $-y$ stay a positive angular distance apart, x and y cannot nearly cancel: $|x + y|$ controls $|x| + |y|$. Hence $|k| = |y| = O(|n|)$, so all admissible lattice points lie in an $O(|n|)$ ball. Counting the containing integer cube gives $O(|n|^d)$ without describing any faces of the translated cone intersection.

3 The all-dimensional theorem

For a conic set $\Gamma \subset \mathbb{R}^d \setminus \{0\}$, write

$$K(\Gamma) = \overline{\Gamma \cap \mathbb{S}^{d-1}}^{\mathbb{S}^{d-1}}.$$

Theorem 1 (Sharp translated-cone count)**.** *Let $d \geq 1$ and let $\Gamma_1, \Gamma_2 \subset \mathbb{R}^d \setminus \{0\}$ be conic sets. Assume*

$$K(\Gamma_1) \cap K(-\Gamma_2) = \varnothing, \tag{2}$$

and put

$$\delta = \operatorname{dist}\big(K(\Gamma_1), K(-\Gamma_2)\big) > 0.$$

Then $N(0) = 0$ and, for every nonzero $n \in \mathbb{Z}^d$,

$$N(n) \leq \left(1 + \frac{4|n|}{\delta}\right)^d \leq \left(1 + \frac{4}{\delta}\right)^d |n|^d. \tag{3}$$

In particular, condition (1) holds with $\gamma = d$. If the two cones are closed relative to $\mathbb{R}^d \setminus \{0\}$, the source's hypothesis $\Gamma_1 \cap (-\Gamma_2) = \varnothing$ implies (2), so its high-dimensional conjecture follows.

Proof. If either cone is empty there is nothing to prove. Otherwise the two sets in (2) are disjoint compact subsets of the sphere, whence $\delta > 0$. Take nonzero $x \in \Gamma_1$ and $y \in \Gamma_2$, and set

$$a = |x|, \quad b = |y|, \quad u = x/a, \quad v = -y/b.$$

Then $|u - v| \geq \delta$, and the elementary identity

$$\begin{aligned} |x + y|^2 &= |au - bv|^2 \\ &= (a-b)^2 + ab|u - v|^2 \\ &\geq (a-b)^2 + ab\delta^2 \geq \frac{\delta^2}{4}(a+b)^2 \end{aligned}$$

holds because $0 < \delta \leq 2$ and $(a-b)^2 + 4ab = (a+b)^2$. Thus

$$|x + y| \geq \frac{\delta}{2}(|x| + |y|). \tag{4}$$

For a point counted by $N(n)$, use $x = n - k$ and $y = k$ in (4). It follows that

$$|k| \leq |x| + |y| \leq \frac{2|n|}{\delta} =: R.$$

The lattice points in the radius-R ball are contained in the cube $[-R, R]^d$, so their number is at most $(2\lfloor R \rfloor + 1)^d \leq (2R + 1)^d$. This is the first inequality in (3); the second uses $|n| \geq 1$. When $n = 0$, an admissible k would give $-k \in \Gamma_1 \cap (-\Gamma_2)$, contradicting (2).

Finally, if the cones are relatively closed, their spherical traces are already compact. Their literal antipodal disjointness is therefore exactly (2). $\square$

Proposition 1 (Exact robust geometric condition)**.** *For arbitrary nonempty conic sets Γ_1, Γ_2, condition (2) is equivalent to the existence of $c > 0$ such that*

$$|x + y| \geq c(|x| + |y|) \quad (x \in \Gamma_1,\ y \in \Gamma_2). \tag{5}$$

One may take $c = \delta/2$.

Proof. The forward implication is (4). Conversely, if the closed spherical traces meet at u, choose unit $u_j \in \Gamma_1$ and unit $v_j \in -\Gamma_2$ tending to u. Applying (5) to $x = u_j$ and $y = -v_j$ gives $|u_j - v_j| \geq 2c$, a contradiction. $\square$

Thus closed-spherical antipodal separation is the full characterization of the uniform no-cancellation mechanism behind the conjectured estimate. It is not asserted to classify exceptional arithmetic behavior of tangent cones.

4 Why literal open-cone disjointness is insufficient

Theorem 2 (Open-cone counterexample)**.** *There are acute open convex cones $\Gamma_1, \Gamma_2 \subset \mathbb{R}^2$ satisfying $\Gamma_1 \cap (-\Gamma_2) = \varnothing$ for which $N(n) = \infty$ at one fixed $n \in \mathbb{Z}^2$. There are full-dimensional open convex pointed examples in every $d \geq 3$ as well.*

Proof. In dimension two set

$$\Gamma_1 = \{(x_1, x_2) : x_1 > 0,\ 0 < x_2 < x_1\}, \qquad \Gamma_2 = \{(x_1, x_2) : x_1 < 0,\ 0 < x_2 < -x_1\}.$$

Both are open convex cones of angular width $\pi/4$. Every point of Γ_1 has positive second coordinate, while every point of $-\Gamma_2$ has negative second coordinate, so the required intersection is empty. Nevertheless, for $n = (0, 2)$ and every integer $m \geq 2$,

$$k_m = (-m, 1) \in \Gamma_2, \qquad n - k_m = (m, 1) \in \Gamma_1.$$

Hence all the distinct k_m are counted and $N(n) = \infty$.

For $d \geq 3$, append the strict linear inequalities

$$|x_j| < \varepsilon x_2 \quad (3 \leq j \leq d)$$

to each displayed cone, with a fixed sufficiently small $\varepsilon > 0$. The resulting sets are full-dimensional open convex pointed cones; the vectors $(\pm m, 1, 0, \ldots, 0)$ give the same infinite family. $\square$

This also shows why passing from disjoint open cones to disjoint closed spherical traces is necessary: the two cones above approach a common antipodal boundary ray, allowing arbitrarily large cancellation.

5 Optimality and consequence for the source

The exponent d is optimal. Take $\Gamma_1 = \Gamma_2$ equal to the open positive orthant and $n = (N, \ldots, N)$, $N \geq 2$. Each $k \in \{1, \ldots, N-1\}^d$ is counted, so

$$N(n) = (N-1)^d \asymp |n|^d.$$

For relatively closed cones, Theorem 1 supplies condition (a) of the source's Theorem 5.5 with the conjectured and best possible value $\gamma = d$. The same conclusion holds more generally under (2). This closes the geometric gap identified in Remark 2 and makes the source's ensuing product corollary available in all dimensions, subject to its remaining analytic hypotheses. The packet does not independently audit those other hypotheses.

6 Verification and novelty limits

The proof uses only compactness of closed subsets of the unit sphere, the displayed exact norm identity, and the elementary integer-cube count. The counterexample is verified symbolically by its strict inequalities for every $m \geq 2$; no numerical approximation is involved. Eight progressive proof and stress-test attempts are recorded in the accompanying attempt log.

The repository's result, solution, attempt, and proof-gap indexes contained no treatment of arXiv:2403.10350. Bounded searches on 13 August 2026 used the exact title, arXiv identifier, exact open-problem phrases, authors, and core terms about translated cone lattice counts. No later resolution was found. The published 2024 Filomat paper retains the open statement, and a 2025 project review repeats it. Since the decisive lemma is elementary geometry, novelty should nevertheless be treated cautiously pending specialist citation review.

Human review should first confirm the source's intended convention for "cone." If cones are relatively closed, Theorem 1 is a direct full affirmative answer. If they are open, Theorem 2 is a counterexample to the literal claim and Theorem 1 gives the sharp statement repair.

References

[1] A. Aksentijević, S. Aleksić, and S. Pilipović, *On the product of periodic distributions. Product in shift-invariant spaces*, arXiv:2403.10350 (2024); Filomat **38** (2024), no. 23, 8011–8021.

Proof. Let

$$(f_1\psi)_{per} = \sum_{k\in\mathbb{Z}^d} a_{1,k}e_k, \quad (f_2\psi)_{per} = \sum_{k\in\mathbb{Z}^d} a_{2,k}e_k.$$

Note, if $x \in \operatorname{supp}\psi$ and $\xi \in (\mathbb{R}^d \setminus \{\mathbf{0}\}) \setminus \Gamma_1$, then $(x,\xi) \notin WF_{s_1}(f_1\psi)$. The same holds for $f_2\psi$. Since $f_1 \in \mathscr{P}'^{\tau_1}$ and $f_2 \in \mathscr{P}'^{\tau_2}$, we know that

$$\sum_{k\in\mathbb{Z}^d\cap\Gamma_1} |a_{1,k}|^2\langle k\rangle^{-2\tau_1} < +\infty, \qquad \sum_{k\in\mathbb{Z}^d\cap\Gamma_2} |a_{2,k}|^2\langle k\rangle^{-2\tau_2} < +\infty. \tag{5.2}$$

Now as in the proof of Theorem 3, we show that $f = (f_1\psi)_{per}(f_2\psi)_{per}$ exists and $f \in \mathscr{P}'$. $\square$

Remark 1. *Theorem 5.5 can be easily transferred to the case when one has several cones Γ_1^i, $i = 1,\ldots,l_1$ (related to f_1) and Γ_2^j, $j = 1,\ldots,l_2$ (related to f_2) so that $\Gamma_1^i \cap \mathbb{Z}^d$ and $\Gamma_2^j \cap \mathbb{Z}^d$ contain index sets for f_1 and f_2, $i = 1,\ldots,l_1$, $j = 1,\ldots,l_2$ which are compatible index sets.*

Remark 2. *We will show below that under the assumption that $\Gamma_1 \cap (-\Gamma_2) = \emptyset$ then condition in a) of Theorem 7 holds in the case $d = 2$ with $\gamma = 2$. Our hypothesis is that condition a) also holds (with $\gamma = d$) for $d \geqslant 3$, but the structure of cones is more complex and we do not have the proof of this hypothesis for $d \geqslant 3$.*

Proof of the assertion in the Remark 2, for $d = 2$. We can assume that cones are acute because if it is not the case, we divide them into finite sets of such cones. So, assume that cones Γ_1 and $-\Gamma_2$ are acute and have empty intersection. By translation with vector $\overrightarrow{(0,0),(n_1,n_2)}$, there are several different positions of cones so that they have different surface of the domain laying between them. It can be equal to zero but the optimal case (maximal number of points with integer coordinates

Figure 1: Remark 2 on page 13 of the arXiv PDF: the dimension-d cone-count hypothesis, followed by the planar discussion.